US Tariffs Hit Germany's Strong Industrial Regions

District exposure, local business tax and regional aid

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Abstract

This paper allocates the output losses of German manufacturing from the US tariffs in force in August 2026 to Germany's 400 districts (Kreise). A multi-sector trade model with input-output links gives the long-run output change of each manufacturing industry; the industry mix of the 38 NUTS 2 regions turns it into a regional shock, and the export ratio of each district's manufacturing allocates the shock within regions. Manufacturing output falls by about EUR 34 billion a year, 1.6 % of the sales of establishments with 20 and more employees, or EUR 38 billion when the 14 districts with suppressed data, among them Wolfsburg and Ingolstadt, are imputed from regional totals. Half of the loss falls on one fifth of districts. These are prosperous, export-oriented industrial locations: GDP per head in the most exposed fifth is more than 60 % above that of the least exposed fifth, and unemployment is lower. About two thirds of the variation in losses per inhabitant reflect how much manufacturing a district has; a uniform shock of the same size would rank districts almost identically (rank correlation 0.90). The tariff structure adds concentration and tilts the loss toward high-earning, export-intensive cities, but it does not shift it toward weak regions. Only 13 % of the loss falls in areas eligible for regional aid under the joint federal-state programme GRW. Local business tax (Gewerbesteuer) is the direct fiscal channel. A district panel for 2009 to 2023 gives an elasticity of Gewerbesteuer to district-specific manufacturing sales of about 0.2, rising to 0.4 in the most industrial third of districts; the resulting tax loss is EUR 0.3 billion a year (95 % band 0.2 to 0.4). Because the tariff shock is national, the elasticity of about one observed in the 2009 recession is at least as relevant; it gives EUR 1.2 billion. Fiscal equalisation and the shared tax-base levy reduce the net loss of municipalities below these gross figures. Summed over 223 labour market regions, which capture commuting, the concentration falls (the most exposed fifth carries 47 % of the loss) but the profile is unchanged. Twenty-six districts combine a high loss with a shrinking population or a high welfare rate, eleven of them outside any aid area. The ranking of districts is robust to eight alternative allocation rules, data choices and tariff scenarios.

Disclaimer. The views expressed are the author's own and do not necessarily reflect those of any institution. All errors are the author's.

JEL classification. F13, F14, R12, H71.

Availability. District maps and tables: https://wfhungerland.github.io/tradepolicy-site/#kreise. Scripts are referred to by number (for example script 74); Appendix B lists them.

1 Introduction

Since 2025 the United States has levied much higher tariffs on imports from the European Union. In August 2026, after the Supreme Court ruling against the tariffs imposed under the International Emergency Economic Powers Act, the remaining measures are mainly Section 232 tariffs on steel, aluminium, copper, cars, trucks, semiconductors and pharmaceuticals, and Section 301 tariffs. Their aggregate cost for Germany is small: in a multi-sector general equilibrium model the tariffs lower German real income by about 0.07 % in the long run (Hungerland 2026). Aggregate numbers hide where the cost falls. That distribution decides which labour markets, municipal budgets and policy instruments the shock reaches.

This paper allocates the manufacturing output loss to Germany's 400 districts and links it to district data on prosperity, labour markets, municipal finances, demography and regional aid. It makes three contributions.

First, it combines a structural trade model with official district statistics in a transparent downscaling, and tests the downscaling against eight alternatives. Second, it estimates how strongly local business tax responds to district-specific changes in manufacturing sales, and uses the estimate to quantify the fiscal channel. Third, it compares the exposure with the map of regional aid areas.

The main result is that the tariff shock is a shock to strong regions. The most exposed fifth of districts is richer, more industrial and younger than the rest, with lower unemployment. The loss correlates with industry employment per working-age inhabitant (Spearman rank correlation 0.67), with GDP per head (0.47) and earnings (0.42), not with unemployment (−0.10). This profile comes mostly from where German manufacturing is located: a decomposition shows that manufacturing sales per inhabitant account for about two thirds of the variation in the loss, the tariff-specific percentage shock for one third. The tariff structure concentrates the loss further on export-intensive cities with high earnings; controlling for density and Land it is unrelated to unemployment. Regional aid in Germany targets other places. The fiscal hit through local business tax is limited, EUR 0.3 to 1.2 billion a year gross, and concentrated in fiscally strong municipalities that receive little from fiscal equalisation. A minority of 26 districts combines high exposure with a second risk, shrinking population or high welfare dependency; they are the natural focus for policy.

2 Related literature

The literature on regional effects of trade shocks has mostly studied import competition. Autor, Dorn and Hanson (2013) show that Chinese import competition depressed employment and wages in exposed US commuting zones for a decade. For Germany, Dauth, Findeisen and Suedekum (2014) find that the rise of China and Eastern Europe was on balance positive: export gains in some regions outweighed import losses in others. Dippel et al. (2022) link import competition in German regions to votes for parties at the right margin.

Export shocks have received less attention. Fajgelbaum et al. (2020) show for the 2018 US tariffs that retaliatory tariffs hit agricultural counties in the United States hardest. An export shock falls where exporting industry is located; in Germany that means the prosperous manufacturing regions that gained from earlier trade integration.

Methodologically, the paper uses a shift-share design. Goldsmith-Pinkham, Sorkin and Swift (2020) and Borusyak, Hull and Jaravel (2022) discuss when such designs identify causal effects. Here the design is used for measurement, not identification: the shifts come from a structural model (Caliendo and Parro 2015), solved in changes (Dekle, Eaton and Kortum 2008), and the shares from official statistics. Associations between exposure and district characteristics are descriptive.

3 Data

The model is calibrated on the OECD inter-country input-output tables for 2022 (OECD 2025), 81 economies and 50 sectors, with US applied tariffs by origin and product from the CEPII database (Guimbard 2026). Regional industry structure comes from Eurostat's structural business statistics by NUTS 2 region (sbs_r_nuts2021, persons employed by NACE division, latest year 2021 to 2023).

District data come from three official sources. The Regionaldatenbank Deutschland, table 42111-04-02-4, gives sales and foreign sales of manufacturing and mining establishments with 20 and more employees by district, 2009 to 2024. Foreign sales are direct sales to customers abroad; deliveries to German exporters count as domestic. Values are suppressed where few establishments would be identifiable. In 2024, 386 of 400 districts are published, with 82.1 million of 83.6 million inhabitants. The 14 suppressed districts include

large car-making locations (Wolfsburg, Ingolstadt, Dingolfing-Landau, Böblingen, Heilbronn city, Emden), so the totals understate the loss; Section 5 imputes them.

INKAR 2025 (BBSR 2025) provides district indicators for 2023 and earlier years: GDP per head, earnings, industry employment, unemployment, population projections to 2045, welfare (SGB II) rates, election results, and Gewerbesteuer receipts and tax capacity per inhabitant since 1995. The BBSR reference file for the regional aid map 2022 to 2027 gives the GRW status of every municipality. Appendix Table A1 lists all sources.

4 Method

4.1 Output change by industry

A Caliendo-Parro model with input-output links gives the long-run change in German output for each of the model's manufacturing sectors when US tariffs move from their level of 1 January 2025 to that of 22 August 2026. Tariff revenue is rebated, trade deficits are held fixed and there is no retaliation in the central scenario. Trade elasticities are sector medians of ProTEE (Fontagné, Guimbard and Orefice 2022), capped at 15. The companion methods paper documents the model, its validation and its sensitivity (Hungerland 2026).

4.2 Regional shock

Each NUTS 2 region r receives the employment-weighted mean of the industry output changes,

$$m_r = \Sigma_s \, (L_{rs} / L_r) \cdot \Delta Y_s,$$

where L_{rs} is employment in manufacturing division s and ΔY_s the model's output change in per cent. Across the 38 German regions m_r ranges from −0.8 % to −2.0 % (median −1.7 %). Regions with metal, car and machinery industries lose most: Arnsberg (−2.0 %), Braunschweig (−2.0 %) and the Saarland (−1.9 %); Hamburg loses least (−0.8 %).

4.3 Allocation to districts

Within a region the industry mix of districts is not published. Districts therefore inherit the mix of their region, and the shock is scaled by the district's export ratio relative to the region's:

$$\Delta Y_k = m_r \cdot q_k / q_r, \quad V_k = -\Delta Y_k \cdot U_k,$$

where q_k is foreign over total sales of district k in 2024, q_r the same for its region, U_k manufacturing sales and V_k the output loss in euro. The scaling assumes that, given the regional mix, the loss is proportional to sales abroad. It keeps regional totals: summed over districts, the loss equals m_r times regional sales. Section 10 tests the assumption.

The approach rests on three simplifications. The district industry mix is that of its region. Foreign sales cover all destinations, so the shock assumes the national destination mix. And the result is a static long-run effect of the first round: it holds employment and capital in place within Germany and has no spillovers between districts through supply chains or commuting.

5 Where the losses fall

German manufacturing loses EUR 34.0 billion of output a year, 1.6 % of the EUR 2,064 billion sales of establishments with 20 and more employees in the published districts. By district, output falls by 0.3 % to 3.5 % (median 1.5 %). In per cent, districts with very high export ratios lead: Jena (−3.5 %), Landshut city (−3.3 %) and Oder-Spree (−3.0 %). Per inhabitant, locations of large plants in the car, chemical and supplier indus-

tries lead: Regensburg city (EUR 3,440), Ludwigshafen (2,900), Rastatt (2,430), Erlangen (2,380) and Schweinfurt city (2,040), against a median of EUR 290 (Figure 1).

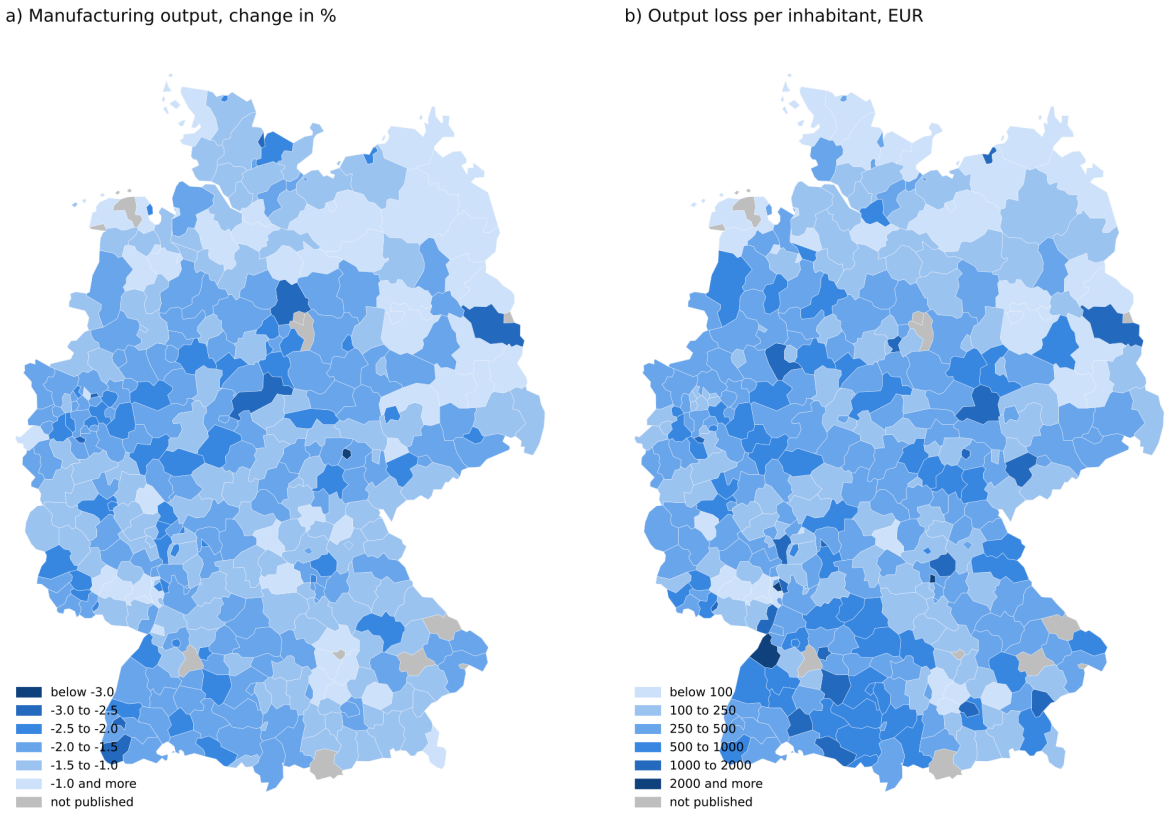


Figure 1: Manufacturing output change and output loss per inhabitant by district, US tariffs of August 2026. Sources: own calculations; Caliendo-Parro model, Eurostat, Regionaldatenbank 42111-04-02-4 (2024). Grey: values not published.

The loss is concentrated. The most exposed fifth of districts carries 50 % of it, the least exposed fifth 3 % (Figure 2). Weighted by population the curve is almost the same, so the concentration does not come from small districts.

The 14 suppressed districts can be estimated. The totals of the Regierungsbezirke (or of the Land) are published, so the suppressed districts of each group are known together as a residual. For eight of them sales are published and only foreign sales are suppressed. Where sales are missing, smaller districts get their gross value added in the secondary sector (national accounts of the Länder, 2022) times the median ratio of sales to that value added in their Land, and the largest plant location takes the rest; foreign sales follow sales times the district's last published export ratio (script 89). With these values the national loss rises to EUR 38.1 billion. The car locations move to the top: Wolfsburg (EUR 13,600 per inhabitant, sales of EUR 83 billion), Ingolstadt (6,300), Dingolfing-Landau (5,700) and Böblingen (3,200). The rule can be tested on published districts (script 89). Masking the largest district of a Regierungsbezirk together with one other district and imputing both gives a median error of 5 % in sales and 8 % in the loss per inhabitant for the large district, and a rank correlation of 0.97 between true and imputed losses. For the small district the median error in the loss per inhabitant is 34 %. Where only foreign sales are missing, as for Böblingen, Emden or Heilbronn city, the median error is below 4 %, although the test uses the export ratio of the previous year while Böblingen's last published ratio dates from 2009. The estimates for Wolfsburg, Ingolstadt and Dingolfing-Landau, the dominant plants of their groups, are therefore reasonably reliable; those for Helmstedt, Garmisch-Partenkirchen and Regen are not. The tables and maps use the published data, and Section 10 treats the imputation as a variant.

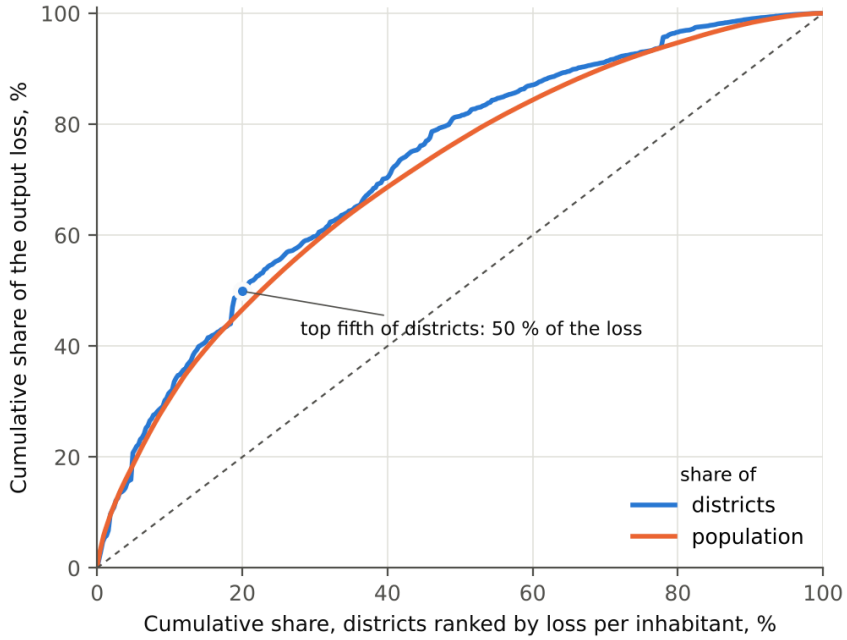


Figure 2: Concentration of the output loss. Districts ranked by loss per inhabitant; blue: share of districts, orange: share of population.

6 Who is exposed

Splitting the districts into five groups of equal size by loss per inhabitant shows a clear profile (Table 1). The most exposed fifth (Q5) has a GDP per head of EUR 61 thousand against 37 thousand in the least exposed fifth (Q1), median earnings 20 % higher, almost twice the industry employment rate, a younger population and lower unemployment. Across all districts the loss correlates most with industry employment per working-age inhabitant (Figure 3).

Table 1: Profile of districts by exposure to US tariffs (quintiles of loss per inhabitant)

Indicator (population-weighted mean)	Q1	Q2	Q3	Q4	Q5
Output loss per inhabitant, EUR	94	192	310	438	919
GDP per head, EUR thousand (2022)	37.2	40.5	44.3	44.7	60.6
Median full-time earnings, EUR per month	3,420	3,683	3,768	3,677	4,104
Industry employees per 100 aged 15-64	12.0	14.2	16.7	20.7	22.6
Employees with university degree, %	14.4	18.9	17.0	15.4	20.9
Unemployment rate, %	6.0	6.2	5.9	5.1	5.5
Mean age, years	45.2	44.3	44.3	44.4	43.5
Population change 2013-2023, %	3.6	3.4	3.3	3.3	3.7

Source: BBSR (2025), INKAR; own calculations. Indicators 2023 unless stated.

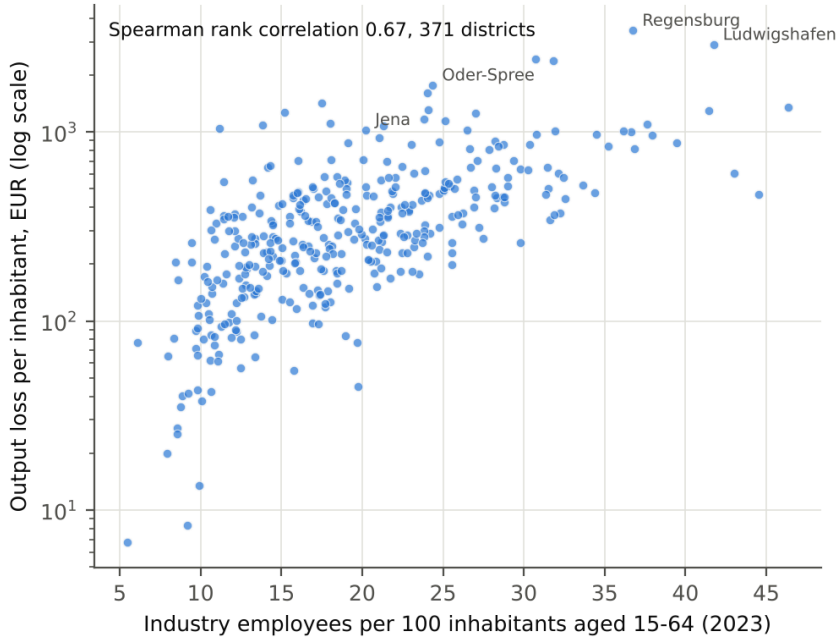


Figure 3: Output loss per inhabitant and industry employment rate by district.

Regressions with Land fixed effects, robust standard errors and standardised regressors (script 79) confirm the pattern. The share of Gewerbesteuer in municipal tax capacity rises by 2.7 percentage points per standard deviation of exposure ($p = 0.007$), controlling for earnings and density.

6.1 What the tariff structure adds

The loss per inhabitant is the product of two factors: manufacturing sales per inhabitant and the percentage output loss, which carries the tariff structure through the regional industry mix and the district export ratio:

$$V_k / P_k = (U_k / P_k) \cdot s_k, \quad s_k = -\Delta Y_k.$$

Any shock proportional to manufacturing output would rank districts by the first factor alone. A benchmark makes this explicit (script 91): a uniform shock that lowers manufacturing output everywhere by the national average of 1.65 % has the same total loss as the tariffs. Its ranking of districts by loss per inhabitant is almost identical to the tariff ranking (rank correlation 0.90). In logarithms, manufacturing intensity accounts for 69 % of the variance of the loss per inhabitant and the tariff-specific percentage shock for 31 %.

The profile of Table 1 therefore mostly describes where German manufacturing is. Figure 6 separates the two factors. Manufacturing intensity carries the strong link with industry employment (rank correlation 0.78) and the negative link with unemployment (−0.20). The tariff-specific shock s_k correlates only weakly with industry employment (0.17), more with earnings (0.38) and GDP per head (0.24), and slightly positively with unemployment (0.18) and the SGB II rate (0.23). The last two links reflect cities: the shock is larger in independent cities with high export ratios (median 1.69 % against 1.46 % in rural districts, rank correlation with density 0.36). With Land fixed effects and density as control, s_k is unrelated to unemployment and the SGB II rate ($p = 0.50$ and 0.97), while one percentage point more output loss goes with 4 % higher earnings ($p < 0.001$). Table 7 shows the raw profile by quintile of s_k.

Table 7: Districts by tariff intensity (quintiles of the percentage output loss)

Quintile	Districts	Output loss, %	Manufacturing sales per inhabitant, EUR thousand	GDP per head, EUR thousand	Median earnings, EUR	Unemployment rate, %	SGB II rate, %	Population in C areas, %
1	78	0.85	24.9	43.4	3,516	6.0	8.2	28
2	77	1.24	18.7	38.5	3,546	4.5	6.1	15
3	77	1.52	27.3	43.5	3,712	4.9	6.9	9
4	77	1.76	20.2	47.9	3,834	6.4	9.8	22
5	77	2.21	34.1	53.7	4,016	6.2	9.3	13

Source: own calculations; BBSR (2025), INKAR. Manufacturing sales of establishments with 20 and more employees, 2024. Means weighted by population.

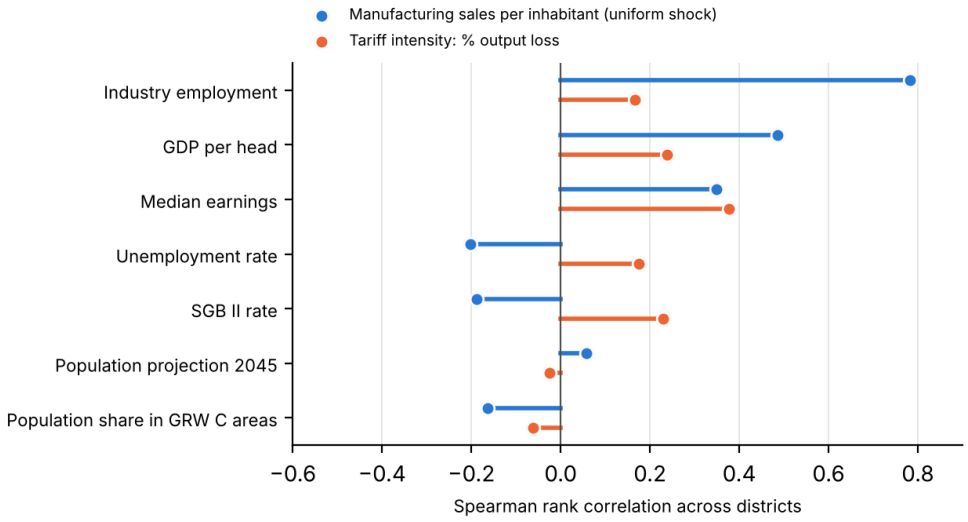


Figure 6: Rank correlations of the two factors of the loss per inhabitant with district characteristics. Blue: manufacturing sales per inhabitant, which alone ranks districts under a uniform shock; orange: the tariff-specific percentage output loss.

What the tariff structure adds is concentration, not a different set of regions. Under the benchmark the most exposed fifth of districts carries 45 % of the loss, under the tariffs 50 %. The share of the loss in GRW C areas hardly changes (13.1 % under the benchmark, 12.5 % under the tariffs). For policy this matters: the mismatch with regional aid is a property of German industrial geography, and it would arise for almost any shock to manufacturing.

6.2 Missing links

Several expected links are absent. Neither the change in the export ratio since 2019 nor the energy intensity of the regional industry mix explains where industry employment fell between 2019 and 2023. Unemployment is slightly lower in exposed districts (5.5 % against 6.0 %), but INKAR has no vacancy totals that would show whether displaced workers would find new jobs locally.

7 Local business tax

7.1 The channel

Gewerbesteuer is the main own tax of German municipalities, EUR 73 billion in 2023. Its base is business profit with add-backs; for firms with plants in several municipalities it is apportioned by payroll (§ 29

GewStG). Prepayments follow the last assessment, so receipts react with a lag. In the most exposed fifth of districts Gewerbesteuer brings EUR 1,269 per inhabitant, twice as much as in the least exposed fifth (EUR 599), and 80 % of tax capacity against 60 %.

7.2 Elasticity

To translate output into tax, the paper estimates the elasticity of Gewerbesteuer per inhabitant to manufacturing sales per inhabitant in a district panel for 2009 to 2023 (script 87):

$$\ln G_{kt} = \beta \ln U_{kt} + \alpha_k + \delta_t + \varepsilon_{kt},$$

with district effects α_k and year effects δ_t, standard errors clustered by district. Year effects remove the national cycle, so β measures the response to a district-specific change in sales, which is the kind of variation the regional allocation of the tariff loss creates. The apportionment of the tax base by payroll is part of this reduced form.

The elasticity is about 0.2 and stable across specifications (Table 2, Figure 4): 0.24 in levels, 0.18 with population weights, 0.17 for the sum of current and two lagged changes in first differences, and 0.25 in long differences between 2010-12 and 2021-23. In the most industrial third of districts it is 0.43, twice that of the other two thirds. With standard errors clustered by NUTS 2 region, which allows for regional shocks, all estimates stay significant; the standard error of the levels estimate rises from 0.047 to 0.054. In a placebo test next year's sales add 0.08 while the current-year coefficient stays at 0.19. The lead is insignificant with district clusters (standard error 0.045) and marginally significant with regional clusters (0.037), which fits persistent sales or anticipated changes; the current-year response is more than twice as large.

Table 2: Elasticity of Gewerbesteuer to manufacturing sales, district panel 2009-2023

Model	Elasticity	Std. error, district clusters	Std. error, NUTS 2 clusters	Observations
Levels, district and year effects	0.239	0.047	0.054	5,777
Levels, population-weighted	0.180	0.049	0.055	5,777
First differences, sum of lags 0 to 2	0.172	0.052	0.058	4,592
Long difference 2010-12 to 2021-23, Land effects	0.249	0.066	0.081	383
Levels with next year's sales: current year	0.191	0.051	0.059	5,384
Levels with next year's sales: next year (placebo)	0.079	0.045	0.037	5,384
Levels, least industrial third	0.205	0.072	0.072	1,913
Levels, middle third	0.173	0.062	0.055	1,980
Levels, most industrial third	0.426	0.123	0.151	1,884

Source: BBSR (2025), INKAR; Regionaldatenbank 42111-04-02-4; own calculations. Standard errors clustered by district (long difference: heteroskedasticity-robust) and, as a check, by NUTS 2 region. First differences exclude changes of more than 150 % in tax and 100 % in sales.

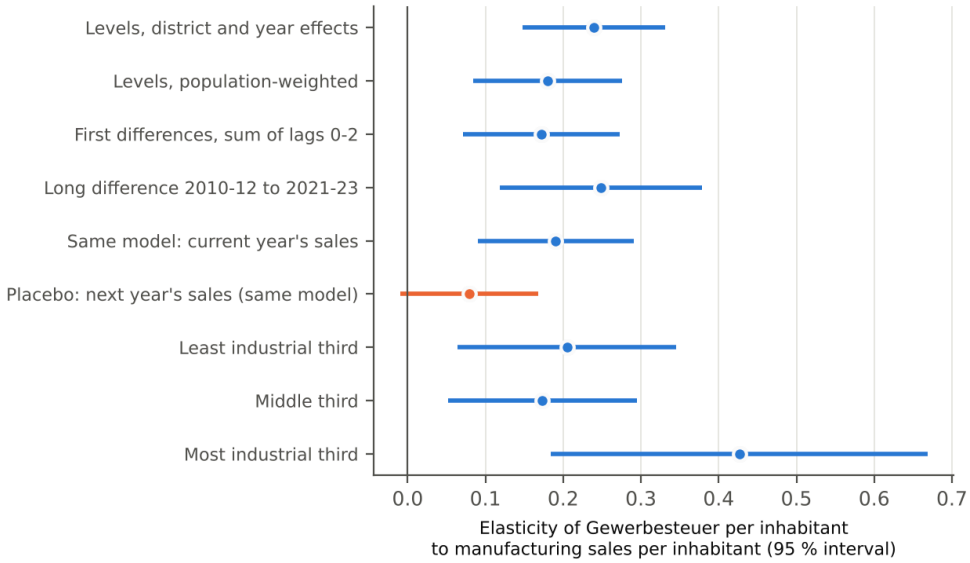


Figure 4: Elasticity of Gewerbesteuer to manufacturing sales, all specifications, 95 % intervals.

The district elasticity is far below the national one. In 2009 Gewerbesteuer fell by 21 % while manufacturing sales fell by about 21 %, an elasticity near one. A national shock lowers profits in all sectors, through demand and supply chains, and through firms' headquarters; a district-specific shock moves only the local plants' share of profits. Because the tariff shock is national, the district elasticity gives a lower bound and the national one an upper bound, and the upper bound is at least as plausible as the lower.

7.3 Tax loss

Applying the tercile elasticities to the district output changes gives a tax loss of EUR 0.30 billion a year, 0.4 % of receipts (95 % band 0.18 to 0.43 billion, from the larger of the district and regional standard errors of the tercile estimates). A uniform elasticity of 0.24 gives 0.29 billion; an elasticity of one gives 1.2 billion. The loss rises from EUR 1.3 per inhabitant in the least to EUR 6.6 in the most exposed fifth (Table 3). With these elasticities even the most affected municipalities lose less than 1 % of their tax capacity, and with an elasticity of one about four times as much: Coburg city (EUR 22 per inhabitant), the district of Munich (20), Erlangen (18) and Regensburg city (17). Places with a high output loss but low current profits, such as Ludwigshafen, lose little tax. Of the tax loss 11 % falls in GRW C areas and 29 % in C or D areas. With the imputed districts the loss is EUR 0.31 billion; Böblingen, Wolfsburg and Dingolfing-Landau then lose EUR 16 to 20 per inhabitant.

These are gross figures; three mechanisms reduce the net loss of municipalities. A share of Gewerbesteuer goes to the federal government and the Länder as a levy on the tax base (Gewerbesteuerumlage), under a tenth of receipts at the usual municipal rates (Hebesätze) above 400 %. Lower tax capacity reduces the contributions that municipalities pay to their district (Kreisumlage) and, in some Länder, to equalisation funds. Municipalities below their fiscal need receive higher general grants (Schlüsselzuweisungen) under the fiscal equalisation of their Land, which replaces a large part of a loss in tax capacity. The exposed municipalities are often fiscally strong and receive few such grants, so they bear more of their loss than weaker places. Quantifying the net effect requires the equalisation rules of each Land and is left for further work.

Table 3: Gewerbesteuer loss by exposure quintile, central elasticities

Quintile	Gewerbesteuer 2023, EUR per inhabitant	Tax loss, EUR per inhabitant	Tax loss, % of Gewerbesteuer	Tax loss, % of tax capacity
1	598	−1.3	−0.21	−0.13
2	725	−2.3	−0.31	−0.20
3	908	−3.3	−0.37	−0.26
4	849	−4.0	−0.49	−0.34
5	1,269	−6.6	−0.57	−0.44

Source: own calculations. Tax capacity (Steuerkraft) as defined by INKAR, 2023.

8 Regional aid and double risks

The GRW, the joint federal-state programme for regional economic structure, supports investment in C areas (eligible under EU state aid rules) and, with lower rates, in D areas. In C areas live 17 % of the population, in C or D areas 42 %. They account for only 13 % and 32 % of the output loss. In the most exposed fifth of districts 8 % of the population lives in C areas, in the least exposed fifth 27 % (Table 4). Of the twelve districts with the highest loss per inhabitant only Oder-Spree is a C area.

Table 4: Population in GRW aid areas by exposure quintile

Quintile	Districts	Population in C areas, %	Population in C or D areas, %
1	78	27	67
2	77	26	46
3	77	17	41
4	77	12	43
5	77	8	23

Source: BBSR, Raumgliederungen Referenz 2023; own calculations. Municipalities with split status counted in equal parts.

On average the shock does not hit weak districts, but some combine a high loss with a second risk. Of the 77 districts in the most exposed fifth, 26 either shrink by more than 5 % to 2045 in the BBSR projection or have an SGB II rate in the top fifth of districts (10.2 % or more); four have both (Appendix Table A2). Two groups stand out: western industrial cities with high welfare rates, such as Ludwigshafen, Leverkusen, Mannheim and Krefeld, and shrinking industrial districts in eastern Germany, the Saarland and the Sauerland, such as Zwickau, Saalekreis, Saarlouis and Olpe. Eight of the 26 are fully or mainly C areas, seven D areas, and eleven lie outside any aid area.

9 Labour market regions

Districts are where plants are, not where all their workers live. Many plants draw commuters from neighbouring districts, so a district-level loss overstates the concentration of income losses. The BBSR groups the 400 districts into 223 labour market regions by commuting flows. Summing the losses (including the imputed districts) over these regions (script 90) gives a median loss of EUR 320 per inhabitant.

The concentration falls but stays high. The most exposed fifth of labour market regions carries 47 % of the loss, against 55 % for the most exposed fifth of districts with the same data; the coefficient of variation of the loss per inhabitant falls from 1.9 to 1.1. The leading regions are Dingolfing-Landau (EUR 5,700 per inhab-

itant), Wolfsburg (2,000, seven districts), Baden-Baden (2,000), Ingolstadt (1,900), Frankfurt (Oder) with Oder-Spree (1,400) and Tuttlingen (1,300) (Figure 5).

Table 6: Labour market regions by exposure to US tariffs (quintiles of loss per inhabitant)

Quintile	Regions	Loss, EUR per inhabitant	GDP per head, EUR thousand	Industry employees per 100 aged 15-64	Unemployment rate, %	Population in C areas, %	Population in C or D areas, %
1	45	130	39.6	11.7	7.5	45	78
2	44	240	39.5	15.5	6.1	23	63
3	45	343	44.8	16.7	5.9	15	41
4	44	472	48.9	18.8	5.4	10	30
5	45	927	55.1	22.3	4.5	6	17

Source: BBSR (2025), INKAR and Raumgliederungen Referenz 2023; own calculations including the imputed districts. Means weighted by population.

The profile is the same as for districts, and in part sharper (Table 6). GDP per head rises from EUR 40 thousand in the least to 55 thousand in the most exposed fifth, unemployment falls from 7.5 % to 4.5 %. The rank correlation of the loss with industry employment is 0.58, with GDP per head 0.54, with earnings 0.50 and with unemployment −0.28. The overlap with regional aid is even smaller: 6 % of the population in the most exposed fifth of regions lives in C areas, 44 % in the least exposed fifth, and 11 % of the loss falls in C areas.

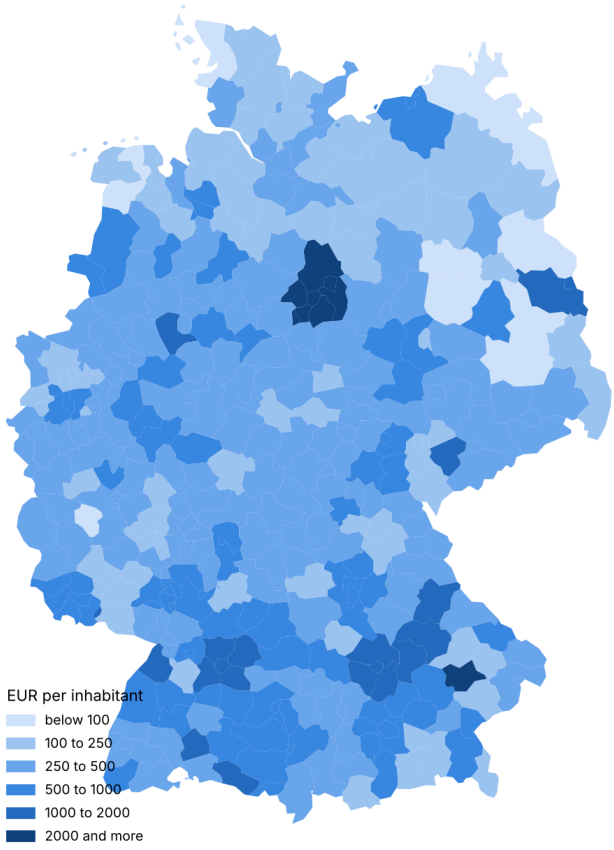


Figure 5: Output loss per inhabitant by labour market region, including imputed districts. Each district shows the value of its labour market region.

10 Robustness

The district allocation rests on assumptions that cannot be tested directly with published data. Eight variants change one assumption at a time (script 86; Table 5):

1. Export ratio and sales of 2019, before Covid and the tariffs.
2. Export ratio and sales averaged over 2022 to 2024.
3. No scaling by the export ratio: every district takes its region's shock.
4. Half of the shock through exports, half proportional to output: $m_r \cdot (0.5 + 0.5\, q_k / q_r)$.
5. The regional loss allocated by industry employment times export ratio instead of sales times export ratio, which removes differences in productivity.
6. The tariff schedule after the framework deals of September 2025.
7. The August 2026 schedule with EU retaliation.
8. The baseline plus the 14 suppressed districts, imputed as in Section 5.

Table 5: Robustness of the district results

Variant	Districts	Loss, EUR bn	Rank corr. with baseline	Top fifth kept, %	Rank corr. with industry employment	Loss in C areas, %	Gewerbesteuer Q5/Q1	Double-risk districts (in baseline)
Baseline (export ratio 2024)	386	34.0	1.00	100	0.67	13	2.12	26 (26)
Export ratio and sales 2019	376	28.9	0.96	83	0.69	12	1.75	25 (22)
Export ratio and sales 2022-2024	380	33.8	0.99	95	0.68	13	1.93	24 (24)
No export ratio (NUTS 2 mix only)	386	34.0	0.92	83	0.78	13	1.84	22 (21)
Half exports, half output	386	34.0	0.98	90	0.74	13	1.83	22 (21)
Employment instead of sales	371	33.6	0.87	69	0.74	13	1.74	18 (14)
Scenario framework deals Sept 2025	386	44.9	0.98	94	0.70	12	1.86	24 (23)
Scenario with EU retaliation	386	28.3	1.00	97	0.69	12	1.86	25 (24)
With imputed values for the 14 suppressed districts	400	38.1	0.99	92	0.68	11	2.16	27 (25)

Source: own calculations. Variants 3 to 5 keep regional totals, so the national loss is unchanged; variants 1, 2 and 5 cover fewer districts because more values are suppressed; variant 8 covers all 400.

The ranking of districts barely changes: the rank correlation with the baseline lies between 0.87 (employment weights) and 1.00 (retaliation). Between 69 % and 97 % of the baseline top fifth stay in the top fifth. The profile holds in every variant: the rank correlation with industry employment lies between 0.67 and 0.78, the share of the loss in C areas between 11 % and 13 %, and Gewerbesteuer per inhabitant in the top fifth is 1.7 to 2.2 times that of the bottom fifth. The double-risk list is less stable: 18 to 27 districts per variant,

14 to 25 of them in the baseline list. Forty-two districts are in the top fifth under all seven sales-based variants with published data (Appendix C).

11 Limitations

The allocation inherits the regional industry mix; within-region variation comes only from the export ratio. A district with a car plant in a machinery region is assigned the machinery region's mix. Employment by industry division and district from the Federal Employment Agency would remove this weakness. The export ratio covers establishments with 20 and more employees, and the 14 suppressed districts include some of the largest car plants; their imputation rests on regional residuals and assumed ratios of sales to value added. Foreign sales cover all destinations, so districts with a high US share are understated and districts selling mainly within the EU overstated. The model gives a long-run, first-round effect without spillovers between districts through supply chains or labour mobility; labour market regions (Section 9) capture commuting, not supplier links. Associations with district characteristics are not causal, and the tax figures are gross of fiscal equalisation. Finally, the double-risk list depends on thresholds.

12 Policy conclusions

The US tariff shock mainly hits economically strong districts outside the regional aid map, largely because German manufacturing is located there. Instruments of regional structural policy therefore reach only a small part of the exposure. Three conclusions follow.

First, the fiscal risk for municipalities should be monitored, but it is limited. The estimated gross loss of Gewerbesteuer is EUR 0.3 to 1.2 billion a year, between 1 % and 4 % of tax capacity in the most affected places, and fiscal equalisation reduces the net loss of weaker municipalities more than that of strong ones. An early warning through the tax estimate and the Länder fits better than new support programmes.

Second, the 26 districts with a double risk deserve separate attention. For the eleven outside any aid area, mainly western industrial cities, the options are network and skills programmes or a change to the aid map for the period from 2028.

Third, the data need improving. The largest gap is the confidential industry mix of districts. A special evaluation of the employment statistics by industry division and district would sharpen the estimates, and the US share of foreign sales by Land would separate US exposure from general export orientation.

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Appendix A: Data

Table A1: Data sources

Source	Content	Use
OECD ICIO 2025	Input-output tables 2022, 81 economies, 50 sectors	Model baseline
CEPII US applied tariffs (Guimbard 2026)	US tariffs by origin, HS6 and date, 2025-2026	Tariff shock
ProTEE 0.1	Trade elasticities by HS6	Sector elasticities
Eurostat sbs_r_nuts2021	Persons employed by NACE division and NUTS 2 region	Regional industry mix
Regionaldatenbank 42111-04-02-4	Sales and foreign sales of manufacturing, establishments 20+, districts, 2009-2024	Export ratio, sales, tax panel
INKAR 2025 (BBSR)	District indicators, 1995-2023, projection to 2045	Profile, Gewerbesteuer, risks
BBSR Raumgliederungen Referenz 2023	GRW status of each municipality	Aid areas
Destatis district list, 31.12.2024; GISCO NUTS 2024	District keys, population, geometries	Matching, maps

Table A2: Districts with high loss and a second risk

District	Loss, EUR per inhabitant	Population 2022-2045, %	SGB II rate, %	Second risk	GRW status	Tax loss, EUR per inhabitant
Ludwigshafen am Rhein (city)	2,902	+4.2	13.5	poverty	none	−5.6
Schweinfurt (city)	2,042	−3.7	10.6	poverty	none	−10.8
Leverkusen (city)	1,424	−1.1	12.3	poverty	none	−6.2
Mannheim (city)	1,314	+3.5	11.3	poverty	none	−12.3
Salzgitter (city)	1,290	−8.1	15.8	shrinking and poverty	none	−4.2
Saalekreis	1,073	−14.2	8.5	shrinking	C	−10.8
Zwickau	1,027	−19.6	7.2	shrinking	C	−5.0
Olpe	962	−9.5	5.2	shrinking	none	−8.7
Holzminden	897	−6.8	10.4	shrinking and poverty	D	−5.8
Pforzheim (city)	875	−0.9	10.5	poverty	none	−7.5
Märkischer Kreis	863	−8.4	9.9	shrinking	D	−7.8
Saarpfalz-Kreis	841	−9.8	7.1	shrinking	D	−6.7
Bremen (city)	826	−2.5	16.3	poverty	D	−3.1
Siegen-Wittgenstein	819	−7.2	8.4	shrinking	none	−9.0
Krefeld (city)	716	−2.7	14.7	poverty	none	−3.1
Saarlouis	713	−10.4	7.7	shrinking	C	−4.5
Remscheid (city)	640	−4.9	12.0	poverty	C	−6.8
Anhalt-Bitterfeld	604	−20.1	10.7	shrinking and poverty	C	−2.3
Tirschenreuth	602	−9.4	3.5	shrinking	C	−10.0
Saalfeld-Rudolstadt	575	−22.0	7.6	shrinking	C	−7.1
Ilm-Kreis	563	−12.2	7.8	shrinking	D	−3.9
St. Wendel	555	−13.2	5.8	shrinking	D	−2.0
Hamburg	555	+7.5	12.2	poverty	none	−3.1
Köln (city)	545	+5.9	13.5	poverty	none	−6.8
Ennepe-Ruhr-Kreis	542	−3.6	10.8	poverty	D	−2.9
Salzlandkreis	535	−20.6	12.4	shrinking and poverty	C	−1.8

Source: own calculations. Second risk: projected population change 2022-2045 below −5 % (shrinking) or SGB II rate of 10.2 % or more (poverty). GRW status by majority of the population; partly: less than half in C or D areas.

Appendix B: Replication

Scripts in the repository, run in this order: 64 (NUTS 2 exposure), 73 (export ratios by district), 74 (district loss), 78 and 79 (INKAR profile and analyses), 80 (GRW), 89 (imputation of suppressed districts), 90 (labour market regions), 86 (robustness), 87 (Gewerbesteuer), 91 (decomposition), 88 (figures); tools/build_regional_paper.py builds this paper. Without the full model run, tools/scenario_from_site.py re-

builds the German sector output changes from the published site data; script 74 then reproduces the EUR 34.0 billion.

Appendix C: Districts in the top fifth under all variants

Alb-Donau-Kreis, Altötting, Amberg, Biberach, Bodenseekreis, Coburg (city), Darmstadt (city), Donau-Ries, Enzkreis, Erlangen, Freudenstadt, Gütersloh, Heilbronn, Hohenlohekreis, Holzminden, Jena (city), Landshut (city), Leverkusen (city), Ludwigshafen am Rhein (city), Mannheim (city), Marburg-Biedenkopf, Märkischer Kreis, München, München (city), Olpe, Ostalbkreis, Pforzheim (city), Rastatt, Regensburg (city), Remscheid (city), Reutlingen, Rottweil, Saarpfalz-Kreis, Salzgitter (city), Schwäbisch Hall, Siegen-Wittgenstein, Soest, Tirschenreuth, Traunstein, Tuttlingen, Unterallgäu, Zwickau.